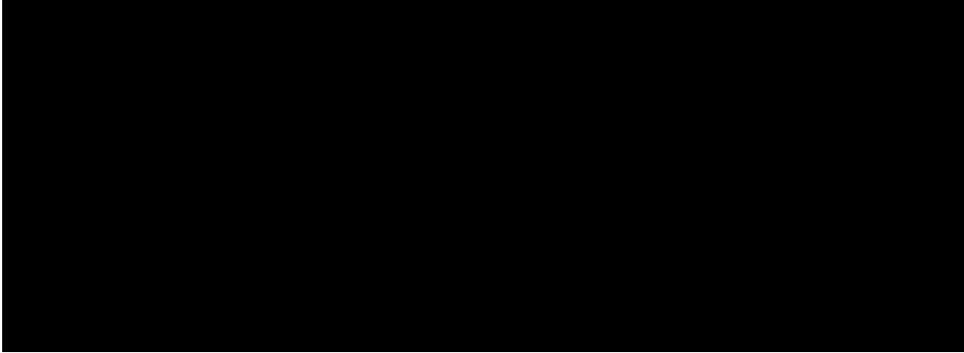


Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Write up a short observing proposal using this template describing your group's final project, what science are you trying to achieve, what telescope/instrument are you using, and how much time do you think you need and why? Work with your group members inside and outside of class to come up with a plan, but you must submit a proposal as an individual. Consult your instructors, TAs, or project leads for help with starting your strategy.



Abstract of Scientific Justification (300 words max) (2 points): Summarize your program. Why is this observing program scientifically interesting? What target(s) are you trying to observe and with what observational modes? What science will your program uncover?

The purpose of this observing program is to constrain the dimensions of an asteroid, and compare its rotational period with archival data. The scientific appeal of this is that we will both observe and calculate the rotational period and light curve amplitude so that we can constrain the elongation of the asteroid. This will help to understand the physical properties of this specific body as well as inform on the properties of similar smaller Solar system bodies as a whole. The program is intended to observe one main-belt asteroid from a list of potential targets, all between 11-15 apparent magnitude. Our main target is 155 Scylla, an asteroid with an apparent magnitude of 14.167. This makes it somewhat dim compared to the others, but its archival rotational period of 7.95 hours makes it a good candidate to track and record its light curve. Our primary backup target is 2 Pallas. We would be imaging this asteroid with the Mont4k CCD mounted to the Kuiper 61" telescope with 60-second exposures and with the I-band filter - in order to gain a lightcurve.

Telescope Requested (0.5 points): *Kuiper 61"*

Instrument (0.5 points): *Mont4K*

Spring 2026A ASTRONOMY 302 Observing Proposal (10 points total)
Add your name to the file name (_LastNameFirstName.docx) then upload to D2L

Tolerable Moon Brightness (dark, grey, bright, etc) (0.5 points): Write out what moon brightness is acceptable for your program and why. How does the moon impact the sensitivity of your observations?

***Bright**, presuming the asteroid isn't passing too close to the moon. Even with a full moon it should be visible through the Mont4k, the asteroids we would be looking at are low magnitude objects (11-15 AP mag) and we would be taking relative brightness measurements.*

Target List (2 points)

List potential targets you may observe and their basic properties like brightness. Under visibility write what months/days the target is up. For example "Object can be observed Feb-March."

REGARDING VISIBILITY: *For asteroids, peak visibility depends entirely on when it hits opposition. This varies by year, so the months listed are current/next window. All of the asteroids listed here were confirmed to be visible in October from the horizons system ephemeris data on the JPL website.

Object Name	Right Ascension	Declination	Apparent V magnitude	Object Type	Visibility
155 Scylla	00 51 54.10	-04 19 27.3	14.167	Asteroid	Aug - Dec
170 Maria	22 50 43.81	+13 26 04.4	13.209	Asteroid	Jul - Nov
188 Menippe	03 02 24.26	+24 23 46.2	13.391	Asteroid	Oct - Feb
167 Urda	01 36 58.47	+07 56 15.8	13.302	Asteroid	Sep - Jan
248 Lameia	02 33 32.76	+18 50 51.8	14.063	Asteroid	June - Oct
295 Theresia	01 39 24.80	+14 59 13.3	13.162	Asteroid	Oct - Feb
2 Pallas	01 19 37.37	-14 59 29.9	8.265	Asteroid	Aug – Dec
236 Honoria	23 05 13.51	-04 08 34.1	11.4	Asteroid	Jul – Nov
71 Niobe	23 12 31.07	+22 53 29.8	11.9	Asteroid	Aug – Nov
695 Bella	23 31 20.56	+24 18 41.0	12.0	Asteroid	Aug - Dec
106 Dione	23 41 02.37	-08 42 08.8	11.8	Asteroid	Aug – Dec
918 Itha	01 16 01.18	+25 32 34.5	14.0	Asteroid	Aug - Dec

Experimental Design & Technical Description (4.5 points total)

How much time is required for your program and what signal-to-noise is necessary for your observations? (2 points)

We would need a SNR of at least around 150 to accurately measure light curves.

For 155 Scylla the rotational period is ~7.95 hours meaning a good baseline for our program time should be at least this amount or a bit more, meaning we will need around $\times 1.5$ this period for a total program duration of ~11.925 hours. Each individual exposure should be kept to about 60 seconds.

2 Pallas has a similar rotational period of 7.81 hours.

How will these observations contribute toward the accomplishment of the goals outlined in the abstract? (2 points)

During the asteroid's rotation period, the relative flux measured should change in a repeated fashion, producing a sinusoidal curve. We can take the average time for the asteroid to return to its initial measured flux over a couple rotational periods, and take that as our final measurement for the rotation period. This observation will produce a scatterplot of time versus flux. We are comparing the flux to the same reference stars with each image to find the relative brightness throughout the rotation. By measuring the changes in this light curve over the full duration, we can further determine the asteroid elongation/shape.

Back Up Plan. Describe a potential project that can be done with poor weather or archival data. (0.5 points)

In the event that we cannot observe, we would attain lightcurve data from online archival data repositories from the Planetary Data System website. We would take data from one of our potential targets and process the data as we otherwise would from there.

<https://sbn.psi.edu/pds/resource/lc.html>

<https://iopscience.iop.org/article/10.1088/1538-3873/ae7cb6/pdf>